Q2 — Frontend Answers

ShelfLife — IA2 Full Stack Web Development

Assumptions

The frontend is designed for the librarian. Members are managed as records and do not need their own login screen. Only the frontend screens required by the assessment are implemented.

Q2(a) — Types and API client

TypeScript interfaces/types are defined for Book, Member and BorrowRecord to match the backend data. A typed fetch-based API client is used for communication with the Express backend.

Q2(b) — Book List

The Book List page fetches books and displays them in a table. It supports searching by title and filtering by genre. Loading and error states are shown while the request is being processed or fails.

Q2(c) — Issue Book

The Issue Book form allows the librarian to select a book and member and provide a due date. It sends the request to POST /api/borrow. A success/error message is displayed and the submit button is disabled while the request is in progress.

Q2(d) — Member History

The Member History page fetches a selected member's BorrowRecord list. Borrow status is displayed using visually distinct badges, including a distinct indication for overdue records.

Q2(e) — Reusable generic component

A generic reusable component was included as part of the frontend implementation. The assessment requirement is to demonstrate reusable typed UI through a generic component such as DataTable or Select.

Q2(f) — Route protection

React Router is used for client-side route protection. If an authentication token is not present, protected pages redirect the user to /login.

State management

Local React state with useState and useEffect is used because the application is small and the required state is page-level. A global state-management library would add unnecessary complexity for this scope.

Implementation scope

Only the frontend tasks explicitly required by the assessment have been implemented. Screens for operations that are only required as backend endpoints, such as adding a book, registering a member, and returning a book, were not added because they are not required frontend screens.